

TORVANE INDUSTRIAL GROUP, INC. —ENGINEERING SERVICES
Kelling Station Unit 2 —Regenerative Thermal Oxidizer Performance Study
EXCERPT —Sections 1, 4 and 6 of 8

Study No.: ES-2013-041

Date: August 19, 2013

Prepared by: Engineering Services, Kelling Station

Approved: W. Hagerman, Vice President, Engineering

Period reviewed: October 2010 –July 2013

1. Purpose and summary

This study quantifies the operating cost impact of the Unit 2 regenerative thermal oxidizer's failure to achieve the performance guarantees of Purchase Order 8-4471, Exhibit C, for use in capital planning and in evaluation of a second media replacement. Total quantified impact through July 31, 2013:

\$3,674,479, comprising excess fuel (\$2,139,879), the April 2012 ceramic media replacement (\$890,000), and unplanned downtime (\$644,600).

4. Excess fuel consumption

Guaranteed natural gas consumption at design load: **14.2 MMBtu/hr**

(PO 8-4471, Exhibit C). Measured average firing rates and computed excess:

Period	Avg firing (MMBtu/hr)	Excess vs. guarantee	Operating hours	Excess fuel (MMBtu)
Oct 2010 –Apr 2012 (pre- replacement, 19 mo)	30.1	15.9	13,110	208,449
May 2012 –Jul 2013 (post- replacement, 15 mo)	21.3	7.1	10,350	73,485
Total			23,460	281,934

The unit fired throughout the extended commissioning and startup period - the oxidizer must be at temperature whenever the lines run, and the lines ran at reduced throughput under the temporary operating permit - so operating hours are taken at the line schedule's 690 hours per month across the full period.

At the weighted average delivered gas price for the period of **\$7.59/MMBtu** (plant meter, invoiced), excess fuel cost is **281,934 × \$7.59 = \$2,139,879**, stated in summaries as approximately \$2,140,000. Operating hours average 690 per month per line schedule.

The April 2012 media replacement improved heat-recovery bed effectiveness from approximately 82% to approximately 89%, against the Exhibit C guarantee of 95%. The unit has at no point achieved the guaranteed fuel consumption.

6. Media replacement and downtime

6.1 Media replacement (April 2012). Full replacement of the structured ceramic heat-recovery media in all three canisters: media supply \$612,000; removal, disposal and installation labor \$198,000; crane and scaffolding \$80,000. Total **\$890,000**. Vendor analysis of the removed media (Appendix D, not excerpted) found thermal cracking and plugging consistent with maldistribution, not with process contamination.

6.2 Unplanned downtime. Eleven days of unplanned Unit 2 line outage in the period are attributable to RTO trips and repairs (outage log reconciliation, Appendix E). At the standard contribution-margin rate of \$58,600 per line-day adopted for FY2013 planning, the downtime impact is **11 × \$58,600 = \$644,600**, stated in summaries as approximately \$645,000. Line throughput reductions taken to preserve permit destruction-efficiency margin are noted qualitatively but are not quantified in this study.

Distribution: W. Hagerman; R. Tenniel; Kelling Station plant manager; Capital Planning file.